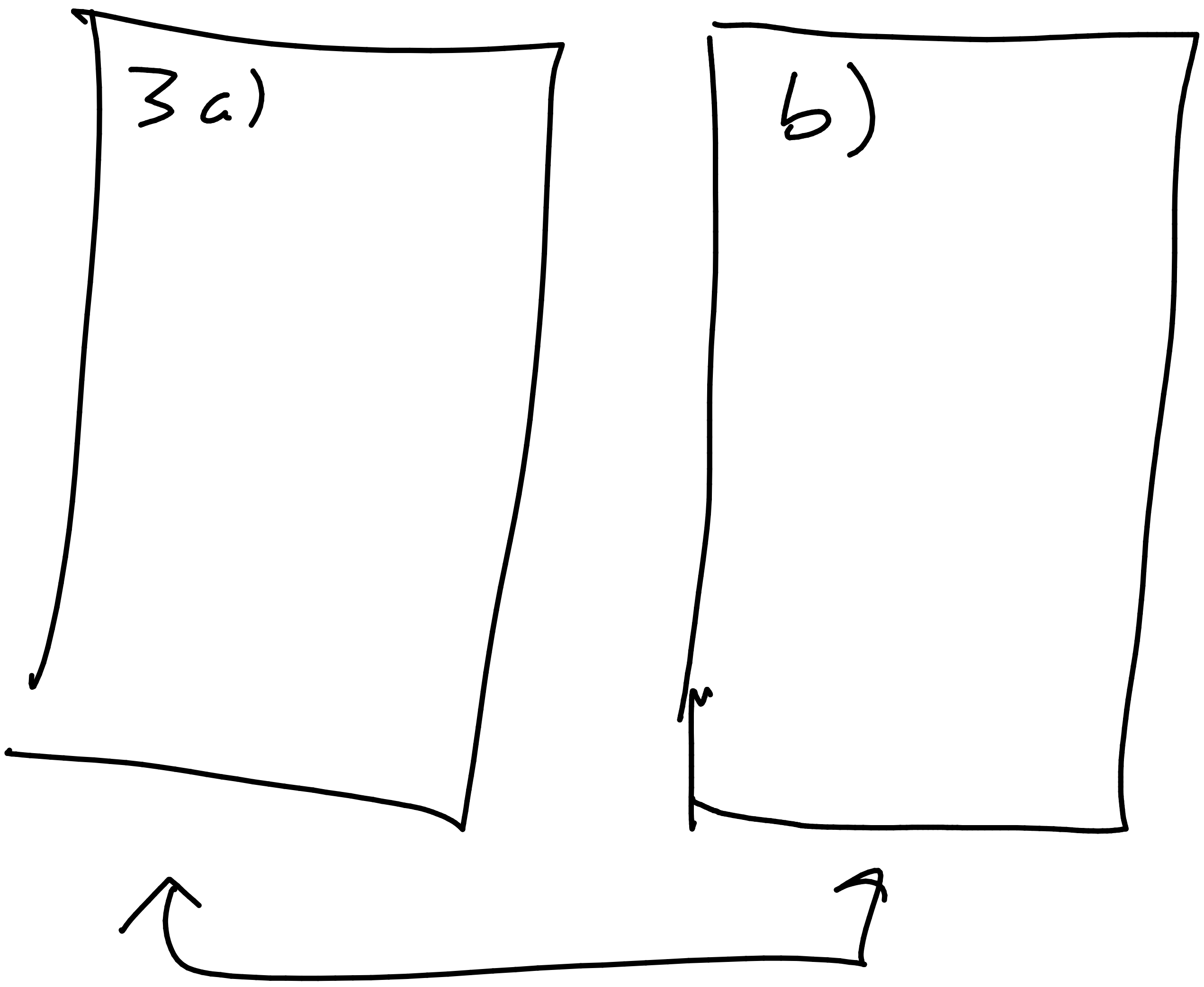
(Todo: pass back old index cards
pass out new blank ones)

Retakes: if a problem is multiple pages, do all of it



problem 3

Time sync in distributed computing

If you have multiple computers, how do you keep their clocks synchronized?

- hopefully down to fractions of a second - milliseconds

Lots of good reasons.

Why is this hard? Just set them the same. Done.

Clocks drift.

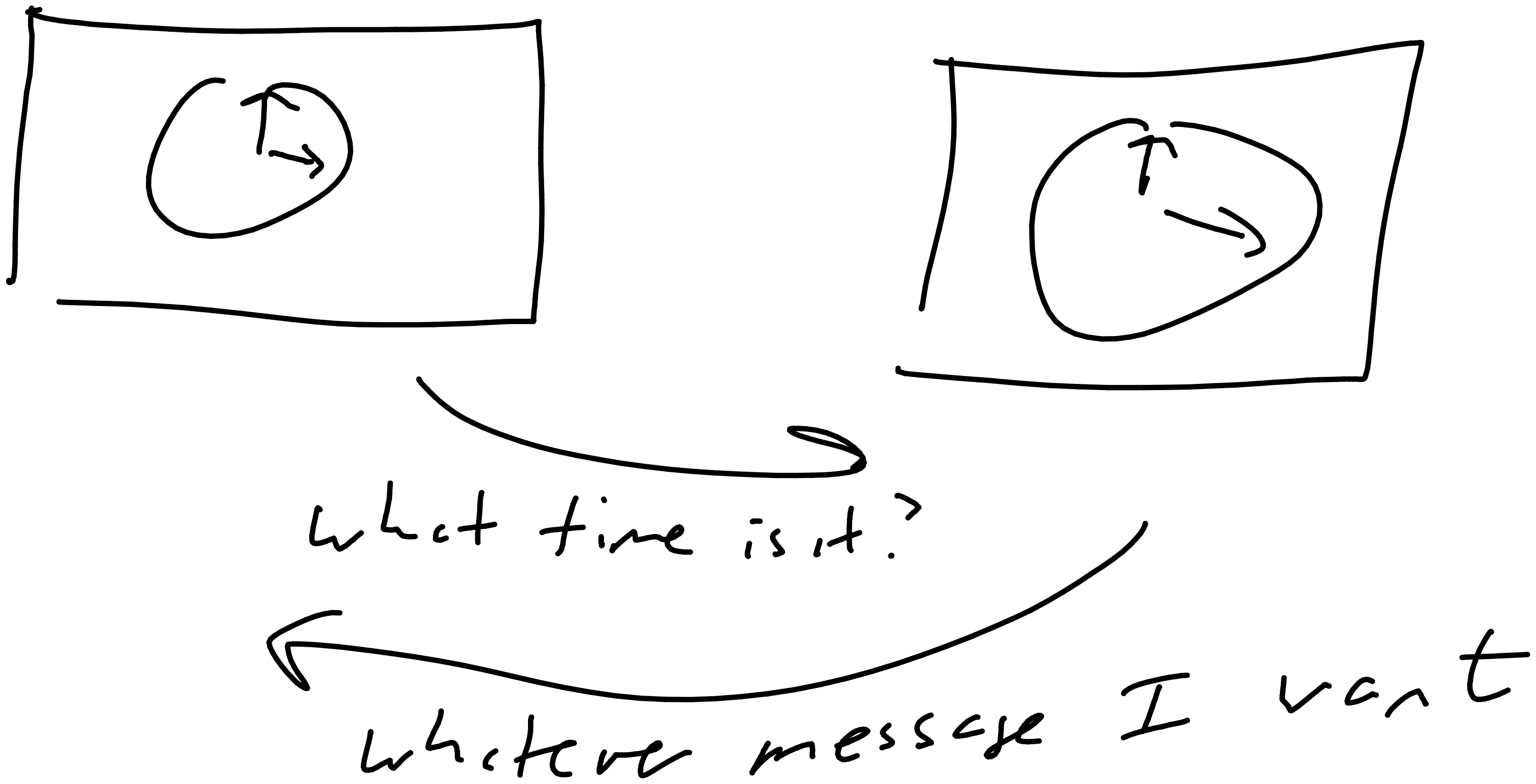
For a typical quartz clock
drift $\simeq 10^{-6}$ seconds, per second
$\rightarrow \simeq 1$ second every 11.6 days

drift can vary, based on temperature, etc

- might not be in same place
(want this automated)
- communication time is real / variable

Thought experiment:

2 computers. They want to figure out how far apart their clocks are. How can they do that?



I don't know how fast or slow communication takes (unless I try to figure it out)

Some variations on how to do this

Simpler version

Computer A sends a message to B,
B sends back to A right away

estimate t = transmission time
as $\dfrac{\text{A receive time} - \text{A start time}}{2}$

More typically
(NTP - network time protocol)

includes t_1

t_1 = msg w/ timestamp (on A's clock)
leaves A

t_2 = msg arrives at B (on B's clock)

t_3 = new msg leaves B (on B's clock)

t_4 = new msg arrives at A (on A's clock)

includes t_1, t_2, t_3

When A gets final msg back, it has all 4 values.

Observations made at A:

$$t_2 = t_1 + \underbrace{t}_{\text{transmission time}} + \theta$$

offset: positive means B's clock is ahead of A

$$t_4 = t_3 + \underbrace{t}_{\text{assume same, yikes}} - \theta$$

Just do algebra and figure out θ

Subtract second eqn from first

$$t_2 - t_4 = t_1 - t_3 + 2\theta$$

$$t_2 - t_1 + t_3 - t_4 = 2\theta$$

$$t_2 - t_1 - (t_4 - t_3) = 2\theta$$

$$\frac{t_2 - t_1 - (t_4 - t_3)}{2} = \theta$$

How do we use this to fix the clocks?

NTP - do this 8 times, throw away outliers, then they average best 3

Who updates? Suppose we find out B's clock is ahead of A.

Do we
- adjust both? (halfway)
- move A ahead?
- move B back?

- Moving clocks back can be really ugly
- Just move A ahead <u>if</u> all you care about is being the same as each other

what if you actually care about correct world time

These are 2 separate problems. If you care about actual world time, you might need to go backwards.

NTP has multiple <u>strata</u> of accuracy.

<u>Stratum 1</u>: Cesium atomic clocks run by governments

various lower strata sync from there